In-orbit measurements of spacecraft microvibrations for satellite laser communication links

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Abstract. Angular microvibrations of platform jitter on the optical inter-orbit communications engineering test satellite are measured in space during ground-to-satellite laser communication links. The microaccelerations are measured by the onboard accelerometers at a sampling rate of 2048 Hz. The angular microvibrations are estimated from the measured microaccelerations using the tracking characteristics of the laser communications terminal and the conversion factor on the basis of microvibration data obtained from ground-based tests. The power spectral density (PSD) of the satellite microvibrations is analyzed by using the fast Fourier transform analysis and the data length is examined according to the frequency resolution of the PSD. The in-orbit measurements of the PSDs are compared with those obtained from the ground test. The angular microvibrational base motion is estimated and a PSD up to 1024 Hz is additionally provided as a database of the real measurement results with previously obtained in-orbit measurements. The measured results will contribute to the angular jitter estimation and the design of a tracking control loop for space laser communication systems in the future. [DOI: 10.1117/1.3482165]

Subject terms: vibration; optical communication; tracking; satellite; laser applications; optical systems.

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1 Introduction

Precision pointing and tracking controls are required for the successful transmission of signals between laser communication terminals in satellite-borne optical communication systems. The optical beam width (of the order of microradians) is a few orders of magnitude narrower than the attitude accuracy of the host satellite, which is usually assumed to be approximately 0.1 deg (1.75 mrad). The accuracy of the satellite attitude can be improved beyond 0.01 deg (0.175 mrad) by using star trackers.[1–3] The strength of the received signal reduces significantly unless special care is taken to reduce the pointing and tracking uncertainties, which are the most serious technological problems affecting satellite-borne optical communication systems.

Satellite microvibrational noise must be analyzed thoroughly for implementing appropriate designs and for the engineering of pointing and tracking systems. The power spectral density (PSD) of microvibrations on the LANDSAT-4 satellite in orbit was first reported by Sudey and Sculman[4] in 1984, and the power spectra of transients caused by the thruster firing systems on the OLYMPUS satellite were reported by Wittig in 1990 (Refs. 5 and 6). The microvibration test conducted on the SPOT-4 satellite was performed on the ground, and the disturbance forces of a SPOT-4 reaction wheel were reported[7] in 1994. Vibration signatures in orbit were obtained using a ground-based CO_2 laser radar while the retractable boom on a satellite was in motion.[8,9] Microvibration tests on the ARTEMIS spacecraft were conducted on the ground, and the in-orbit laser pointing was analyzed[10,11] using Monte Carlo simulation in 1997. A real in-orbit measurement of microvibrations was conducted by using laser communication equipment onboard the ETS-VI satellite.[12,13] Vibration effects and possible solutions for satellite optical communication networks were presented to overcome the problems caused by transmitter pointing vibrations.[14] The results of the on-ground tests and the in-orbit measurements were compared by using the SPOT-4 satellite in 1999 after its launch.[15] The maximum measured acceleration level was approximately 1 mG near the X-wheel. The noise generated due to the nominal rotation speed of the wheels increased the environmental noise at low frequencies. The characteristics of microvibrations on the optical inter-orbit communications engineering test satellite (OICETS) were reported on the basis of data obtained from the ground environment test.[16,17] The PSD for the worst case was predicted.[18] The in-orbit microvibration level of approximately 10 mG was continuously excited along each axis, and it was observed that during the gimbal slewing time of the laser terminal, higher vibration levels were excited on each axis as compared to any other time.[19] The data obtained indicated that the microvibration disturbance level measured in orbit was higher than those obtained for the ground tests, and it was con-

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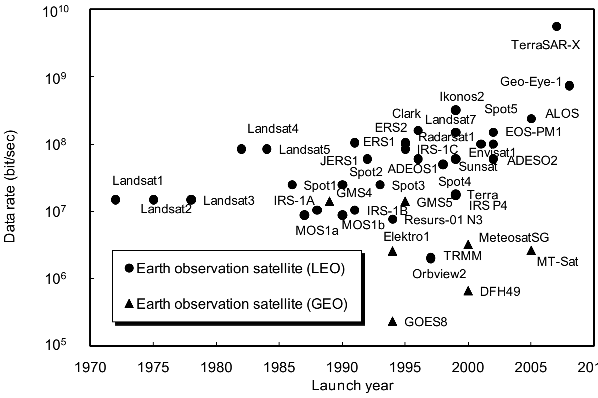

Fig. 1 Data rates for Earth observation satellites.

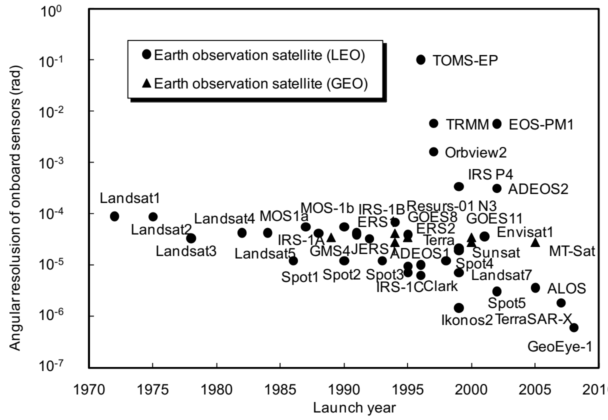

Fig. 2 Angular resolution of sensors on-board Earth observation satellites.

firmed that the ground test results for each axis indicated similar characteristics in orbit in the frequency range between 40 and 1000 Hz.

The objective of the paper is to show the real measurement data of the microvibration base motion on the satellite conducted in a ground test and in space, and to compare the PSDs measured in the ground test and in space based on the angular disturbances during the slewing and tracking modes of the optical antenna. The reminder of the paper is organized as follows. The angular resolution requirements for observation satellites are introduced in Sec. 2, and the equations to calculate the angular microvibration are provided in Sec. 3. The effect of the microvibration to free-space laser communication links is derived in Sec. 4. Section 5 describes the configuration of the microvibration measurements and the specifications of the onboard accelerometers. Section 6 provides the microacceleration levels measured by the onboard accelerometers oriented along three directions. The PSDs of the accelerations are analyzed using the fast Fourier transform (FFT) and the PSDs of the angular vibrations during the laser communication experiments are given based on the derived equations. The PSD obtained in the ground test is compared with that in the in-orbit measurement.

2 Resolution Requirements for Earth Observation Satellites

Earth observation satellites require high-accuracy pointing and stringent line-of-sight stability as well as laser communication terminals. Worldwide high-resolution imagery at the 1- to 3-m resolution, which is available commercially, for example, through Space Imaging, Inc., as a result of declassification and deregulation by the Department of Defense of the United States, will generate massive amounts of data, particularly if the images are multispectral. Either the collected data must be stored and processed on-board the observing satellite to reduce the required downlink transmission rates or high-speed communication links that transmit directly to the ground or via some intermediate relay satellites must be employed. Laser satellite communication can play a pivotal role in the architecture of such systems.

Figures 1 and 2 show the data rates for Earth observation satellites and the resolution of the onboard optical or microwave image sensors.[20] From these figures, it is observed that the data-rate requirements for Earth observation satellites are predicted to grow beyond 10 Gbits/s in the next decade, and the angular resolution of the onboard sensors will improve beyond 1 μrad. Lower vibrational disturbance platforms are required to realize such high accuracies, making it necessary to understand the characteristics of microvibration to ensure the highest pointing accuracy.

3 Microvibrational Disturbance

Recently, the characterization of the transfer functions of a structure with respect to the launch vehicle environments has become a well-recognized field with highly advanced data analysis techniques. When considering microvibrational levels, problems involvie a wider frequency spectrum as compared to that in classical structural modal. Even if the finite element model is fine-tuned after performing the classical structure modal test, many uncertainties remain with respect to microvibration of the satellite in orbit. The spacecraft attitude and orbit control system (AOCS) can compensate for low-frequency disturbances (typically below 1 Hz), but they are not effective at higher frequencies. These high-frequency disturbances may have a significant impact on the spacecraft pointing stability (typically ranging from 20 to 200 Hz); therefore, it is necessary to determine their characteristics.

The microvibration is defined as the angular vibrations beyond around an order of a hertz, and generated from mechanical moving devices on a satellite, which propagates along the structural panels and disturbs payloads or sensors that require high pointing accuracy. A sinusoidal acceleration at the interface point of the payload in a satellite, which occurs due to microvibrations generated from a mechanical moving source k and which is sometimes induced and amplified by a structural resonance, is expressed by

$$a_k(t,f) = A_k(f) T_k(f) \exp\left(j2\pi ft - \frac{x_k}{d} \right), \tag{1}$$

where $A_k(f)$ denotes the amplitude of the microvibration at frequency f on the disturbing device, $T_k(f)$ is the transfer function due to the structure, d is the average damping constant of the microvibration along the propagation path in the satellite, and x_k is the distance from the disturbing source to the interface point. The transfer functions of $T_k(f)$ were measured and shown in Figs. 13 and 14 of Ref. 18, for example. The variance of the acceleration at the frequency f is defined as

$$a_{k,\mathrm{rms}}^2(f) = \overline{[a_k(t,f) - \overline{a_k(t,f)}]^2} = \frac{A_k^2(f)}{2} T_k^2(f) \exp\left(-2\frac{x_k}{d}\right). \quad (2)$$

The microvibrational level at the interface point generated from all the disturbing sources can be expressed as the sum of the variances because these quantities from all the sources are random variables independent of each other. Thus,

$$a_{\mathrm{mount}}^2(f) = \sum_{k=1}^{N} a_{k,\mathrm{rms}}^2(f) = \sum_{k=1}^{N} \left[\frac{A_k^2(f) T_k^2(f)}{2} \exp\left(-2\frac{x_k}{d}\right)\right], \quad (3)$$

which corresponds to the PSD of the microvibration at the interface point of the payload, where N denotes the number of disturbing sources in the satellite. It is apparent that the term $T_k^2(f)\exp(-2x_k/d)$ is the transfer function from the vibrating source k. The damping constant d is experimentally given by 0.574 m for the OICETS satellite, which is the $1/e$ damping length of the amplitude of the accelerations.[18] The damping constant d=0.574 m will be a similar value for general systems, which have the aluminum honeycomb structure. The acceleration level at any point can be predicted by using Eq. (3) and d=0.574 m; hence, it is necessary to give proper consideration to the effects of microvibrations, particularly during the preliminary design phase of the satellite.

The microvibrational displacement causes angular microvibration, and the PSD of the angular microvibration is proportional to that of the displacement. The acceleration can be converted into the angular displacement by using integration twice as expressed by[18]

$$\theta_{\mathrm{rms}}(f) = \frac{C}{(2\pi f)^2} a_{\mathrm{mount}}(f), \quad (4)$$

where C is a constant expressed in unit of inverse meters and experimentally determined, and $\theta_{\mathrm{rms}}(f)$ denotes the angular disturbance at the mounting point of the laser communication terminal. The closed-loop feedback gain $G(f)$ of the optical tracking servo system can reduce the angular disturbance. The residual tracking error in the laser communication terminal passing through the optical tracking servo system is expressed by

$$\theta_{\mathrm{error}}(f) = \frac{\theta_{\mathrm{rms}}(f)}{1+G(f)} = \frac{C a_{\mathrm{mount}}(f)}{4\pi^2 f^2 [1+G(f)]}, \quad (5)$$

where $G(f)$ denotes the rejection gain of the angular perturbation of the optical tracking system at the interface point of the payload. The rejection gain is a function of the frequency and is designed according to the servo control design. The residual angular tracking errors in the laser communication terminal can be obtained by using

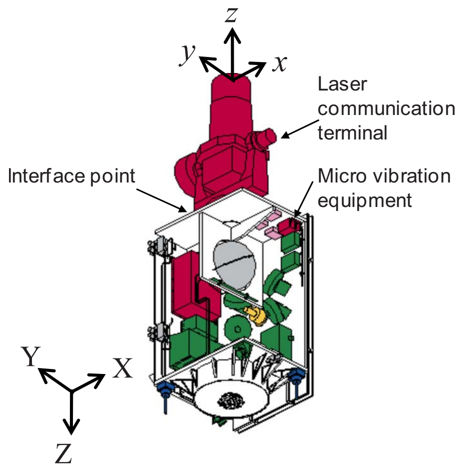

Fig. 3 Definition of the axes in the laser communication terminal and the bus system of the OICETS satellite.

$$\begin{cases} \theta_{\mathrm{error},x}(f) = \dfrac{C_x a_{\mathrm{mount},x}(f)}{4\pi^2 f^2 [1+G(f)]} \\ \theta_{\mathrm{error},y}(f) = \dfrac{C_y a_{\mathrm{mount},y}(f)}{4\pi^2 f^2 [1+G(f)]}, \end{cases} \quad (6)$$

which have units of rad/Hz$^{1/2}$, and $a_{\mathrm{mount},x}(f)$ and $a_{\mathrm{mount},y}(f)$ correspond to the microaccelerations on the interface point that influence the x and y tracking directions, respectively. The values of constants C_x and C_y are usually difficult to determine, however, they can be experimentally determined from the ground test results as[18]

$$\begin{cases} C_x = 51.96\ \mathrm{m}^{-1} \\ C_y = 13.32\ \mathrm{m}^{-1}. \end{cases} \quad (7)$$

In the ground test, the tracking errors and the accelerations were simultaneously measured and then the coefficients of C_x and C_y were estimated. These values of C_x and C_y can be used for converting acceleration to angular vibration, and it enables us to estimate the in-orbit angular microvibrations. The pointing direction (x,y) of the laser communication terminal and the coordinate system (X,Y,Z) of the OICETS satellite bus are defined in Fig. 3. The azimuth angle of the pointing direction is the rotational angle along $-Z$ axis of the satellite bus system and the elevation axis is the rotational angle along the x axis of the laser communication terminal. The interface point means the connecting part between the laser communication terminal and the satellite bus. The difference between C_x and C_y might depend on the direction of the antenna direction and the transfer function, and the coefficients might be different from the designed directions. However, we could not measure C_x and C_y for the other directions on the ground test.

4 Effect of Microvibration to Free Space Laser Communication Links

For definiteness we consider a Gaussian shaped laser beam of $1/e^2$ half width divergence angle w_0 and transmitted power P_t (Refs. 21 and 22). In the absence of jitter, the irradiance distribution of wavelength λ at propagation distance R in the far field at angular coordinate θ can be expressed as[23]

$$I^0(\theta,R) = \frac{P_t\tau_t}{R^2}\frac{2}{\pi w_0^2}\exp\left(-2\frac{\theta^2}{w_0^2}\right), \tag{8}$$

where τ_t is the optics loss of the transmitter. The received optical power at the receiver is given by

$$P_r = I^0(\theta,R)A_r\tau_t, \tag{9}$$

where A_r is the area of the received aperture, and τ_r is the optics loss of the receiver. The received optical power stimulates the photocurrent in the photodetector of which the signal-to-noise ratio (SNR) can be represented by the parameter Q. The bit error ratio (BER) of the optical receiver for the intensity modulation direct detection (IMDD) system with non-return-to-zero (NRZ) is given by[24]

$$\mathrm{BER}(Q) = \frac{1}{2}\,\mathrm{erfc}\left(\frac{Q}{\sqrt{2}}\right), \tag{10}$$

where erfc( ) is the complementary error function. As the SNR can be considered as constant in the absence of jitter, the average BER is directly given by Eq. (10).

The pointing errors have a significant impact on the link performance and bit error probability.[22] The pointing probability density function (pdf) with the root mean square (rms) random jitter $\sigma=[\int\theta_{\mathrm{error}}^2(f)df]^{1/2}$ becomes the Nakagami-Rice distribution and is given by[25–27]

$$p_j(\varphi,\theta) = \frac{\varphi}{\sigma^2}\exp\left(-\frac{\theta^2+\varphi^2}{2\sigma^2}\right)I_0\left(\frac{\varphi\theta}{\sigma^2}\right), \tag{11}$$

where θ is the random pointing jitter angle, φ is the bias error angle from the center of the Gaussian beam, and I_0 is the zero-order modified Bessel function. It is important that the random jitter σ depends on the time span to which one pays attention. The larger jitter should be assumed to evaluate the long-term stability such as for the commercial satellites.[13,28,29] The bias error angle can be even regarded as the zero mean random value because the optical axis of the transmitted laser beam will be calibrated periodically by the calibration process of the optical system during such a long timespan. We assume that the bias error angle can be regarded as zero in this paper, the pdf of the received optical intensity $p(I)$ is given by[25,27,30,31]

$$p(I) = \beta I^{\beta-1} \quad \text{for } 0 \leqslant I \leqslant 1, \tag{12}$$

$$\bar{I} = \frac{\beta}{\beta+1}, \tag{13}$$

where I is the normalized intensity, $\bar{I}$ is the average value, and $\beta=w_0^2/4\sigma^2$. Therefore the unconditional BER in the presence of jitter should be averaged with respect to the pdf of the received optical intensity and is given by

$$\begin{aligned}\overline{\mathrm{BER}}(Q_r) &= \int_0^1 p(I)\mathrm{BER}\left(IQ_r\frac{\beta+1}{\beta}\right)\mathrm{d}I\\ &= \frac{Q_r(\beta+1)}{2}\int_0^1 I^{\beta-1}\,\mathrm{erfc}\left(\frac{IQ_r}{\sqrt{2}}\frac{\beta+1}{\beta}\right)\mathrm{d}I,\end{aligned} \tag{14}$$

where Q_r is the required SNR for the desired average BER, which equals a. The power penalty for average BER due to the variation of the received optical signal is defined as

$$L_j = \frac{Q|_{\mathrm{BER}(Q)=a}}{Q_r|_{\overline{\mathrm{BER}}(Q_r)=a}}. \tag{15}$$

Then, one can calculate the degradation of BER in the optical links due to microvibrational jitter by using Eqs. (14) and (15) if the beam divergence angle and the rms random jitter are given. The impact of the microvibration on BER was examined and there is an optimum value between the beam divergence angle and the rms random jitter.[32]

In addition, if there is atmospheric turbulence under weak turbulence conditions, the pdf becomes the joint function. The pdf of the received optical intensity due to atmospheric turbulence becomes the lognormal distribution, and the pdf due to random pointing jitter is the beta distribution. The total pdf $p(I)$ becomes the joint pdf in the presence of both atmospheric turbulence and random jitter, which is given by[25]

$$\begin{aligned}p(I) = \frac{1}{2\bar{I}}\,\mathrm{erfc}&\left[\frac{\ln(I/\bar{I})+\sigma_I^2(\beta+1/2)}{\sqrt{2}\sigma_I}\right]\\ &\times\beta\exp\left[\frac{\sigma_I^2}{2}\beta(\beta+1)\right]\left(\frac{I}{\bar{I}}\right)^{\beta-1},\end{aligned} \tag{16}$$

where σ_I^2 is the normalized variance called the scintillation index. Therefore, if substituting Eq. (16) into Eq. (14), one can also calculate the degradation of BER in the presence of both atmospheric turbulence and random jitter.

5 Experimental Setup for Microvibration Tests

5.1 *Configuration of the Experiments*

Bidirectional ground-to-satellite laser communication experiments were performed[33,34] in March, May, and September 2006. The laser links were established between an optical ground station and a low Earth orbit (LEO) satellite; the ground station was operated by the National Institute of Information and Communications Technology (NICT, formerly CRL) and located in Koganei, Tokyo, Japan, and the LEO satellite was at an altitude of 610.0 km and inclination of 97.8 deg (OICETS, called "Kirari"). The Kirari Optical Communication Demonstration Experiments with the NICT optical ground station (KODEN) were conducted as a cooperative effort between the Japan Aerospace Exploration Agency (JAXA) and NICT. The ground-to-OICETS laser communication experiments were the first in-orbit demonstration with respect to LEO satellites. Figure 4 shows the configuration of the KODEN experiments. The satellite is controlled from the JAXA's Kirari operation center situated in Tsukuba, Ibaraki, Japan. The optical communication experiment on the link between OICETS and the

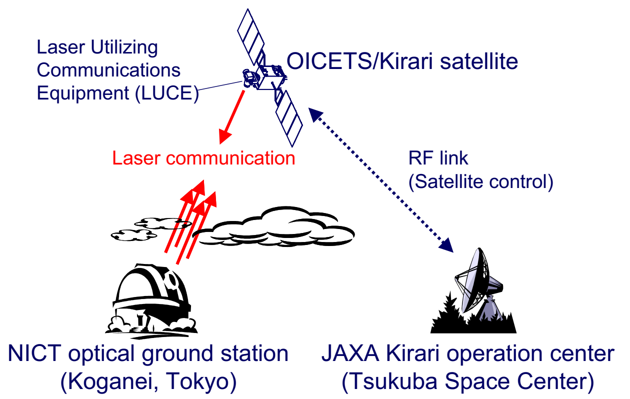

Fig. 4 Configuration of the ground-to-satellite laser communication experiment.

NICT optical ground station was conducted while the satellite was visible from the NICT optical ground station in Koganei, Japan, above an elevation angle of 15 deg. A 1.5-m telescope was used for the uplink transmission.

5.2 *Specifications of the Onboard Accelerometers*

The microvibrations were measured by using accelerometers called the microvibration equipment (MVE) onboard the OICETS satellite, as shown in Fig. 3. The specifications of the MVE are provided[19] in Table 1. The MVE is installed on the top panel of the satellite bus structure, which is the same panel on which the laser terminal is mounted. Microvibration measurements were made with a sensitivity of better than 50 μG and a dynamic range of 100 mG.

6 Results of Microvibration Tests

6.1 *Acceleration Measurements*

Figure 5 shows an example of the accelerations measured by the MVE on September 19, 2006, during the ground-to-satellite laser communications experiments. The maximum acceleration level of 0.5 m/s^2 (=50 mG) was measured for the Z axis during the slewing operation of the telescope. The definition of the axes (X,Y,Z) in the satellite bus system is given in Fig. 3. The gimbal angles of the laser terminal measured during the same period represented in Fig. 5 are shown in Fig. 6.

Table 1 Specifications of the MVE onboard the OICETS.

Item	Value
Type	Servo type
Number of accelerometers	3 for X, Y, Z axes
Frequency range (half power point)	0.5 to 1000 Hz
Full-scale dynamic range	±100 mG
Resolution	better than 50 μG
Sampling rate	2048 Hz
Characteristic frequency	greater than 150 Hz

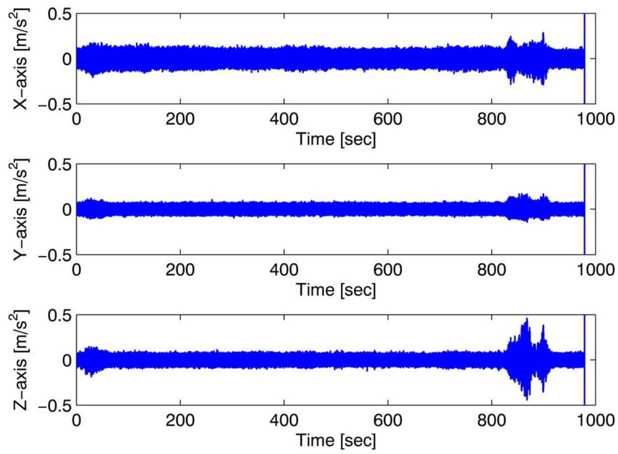

Fig. 5 Typical acceleration responses along the X, Y, and Z axes measured by the MVE onboard the OICETS.

6.2 *PSD of the Acceleration*

Typical frequency responses of the acceleration along the X, Y, and Z axes are shown in Figs. 7(a)–7(c), respectively. The frequency spectrum is calculated at every 1 s and plotted as a function of the experiment time. Figures 8(a) and 8(b) show the frequency spectra of the microvibrational accelerations for the tracking and slewing operations of the optical antenna, respectively, as measured on September 19, 2006. During the tracking operation of the telescope gimbals, some frequency components of the order of approximately a few tenths of hertz are excited. The OICETS satellite has a natural frequency of approximately 30 Hz as the first mode of the structure when in-orbit; this is consistent with the predicted frequency. During the slewing operation, wider frequency components from a few to a few hundred hertz are excited.

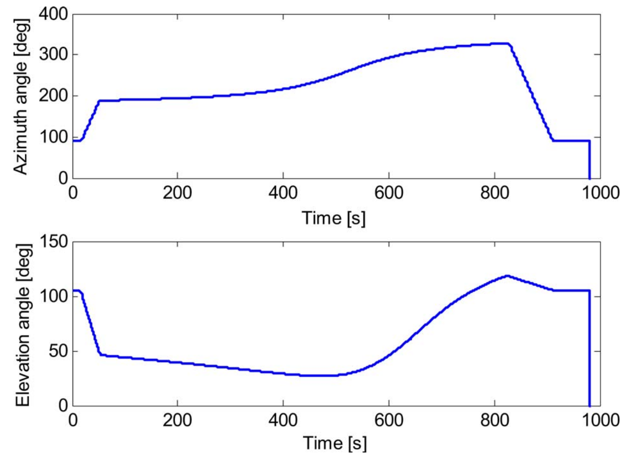

Fig. 6 Typical gimbal angles of the laser communications terminal onboard the OICETS.

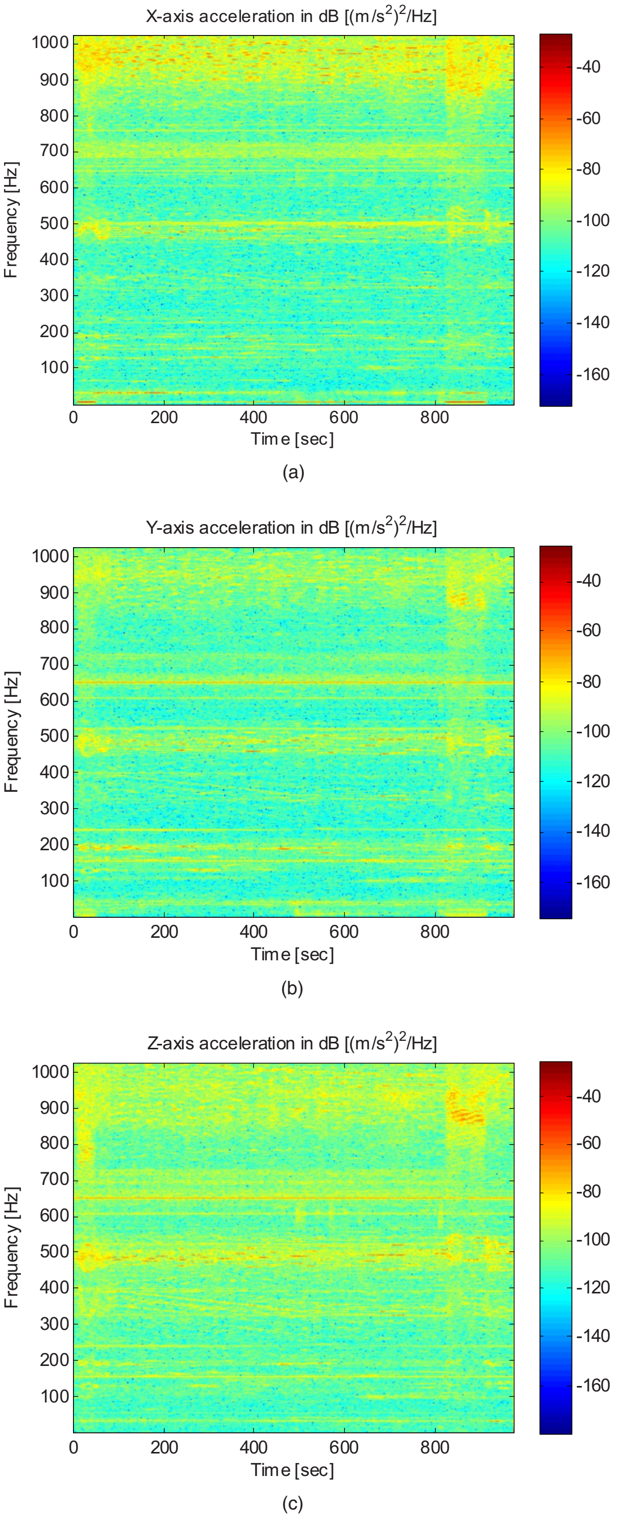

Fig. 7 Typical frequency spectra of the acceleration, as measured on September 19, 2006, along the (a) *X* axis, (b) *Y* axis, and (c) *Z* axis.

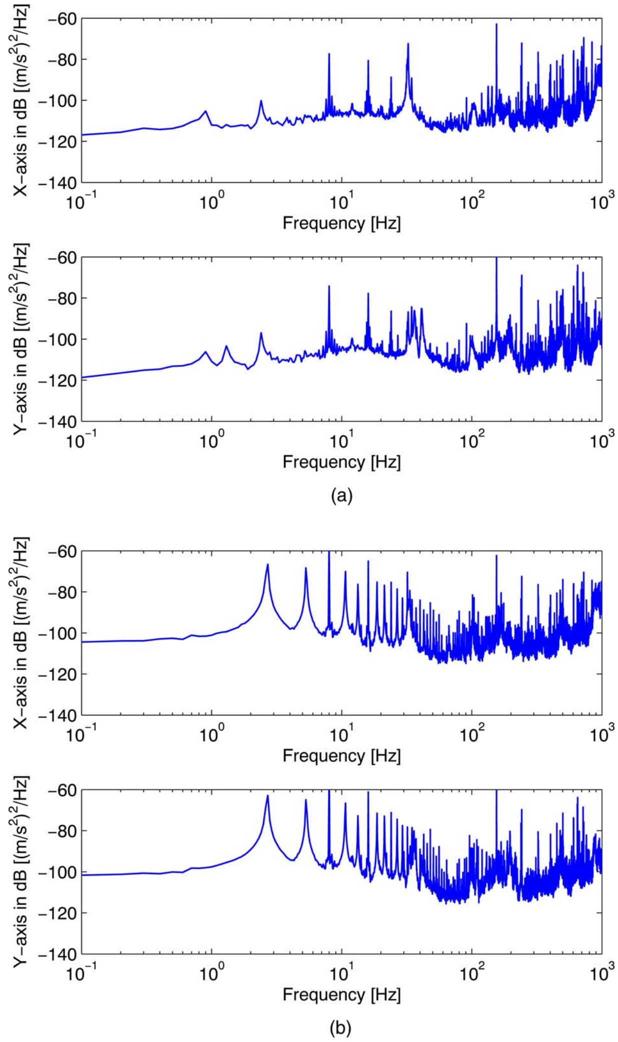

Fig. 8 Frequency spectra of the accelerations while the optical antenna is performing the (a) tracking and (b) slewing operations, as measured on September 19, 2006.

6.3 *PSD of the Angular Microvibration*

Figures 9(a) and 9(b) show the PSD of the angular microvibration levels measured by the MVE during the tracking and slewing modes, respectively, for four successful ground-to-satellite experiment days. From Eq. (4), which provides the relationship between the angular disturbance and the acceleration level, we can estimate the PSD of the angular disturbance at the mounting point of the laser communication terminal [the results are shown in Figs. 9(a) and 9(b)]. The PSD basically falls with frequency by f^4. Figure 10 shows a comparison between the PSDs for the tracking and slewing modes, which are averaged over the four measured data shown in Figs. 9(a) and 9(b). The angular microvibration for the slewing mode is higher than that for the tracking mode at lower frequencies. Frequency components of the order of a few hundred hertz are excited during the optical antenna tracking mode. On the other hand, frequency components of the order of a few hertz are exited during the slewing mode because of mainly the structural responses of the satellite.

6.4 *Comparison of the PSDs*

Figure 11 shows a comparison between the PSDs during tracking measured in the ground test and in space. The in-orbit measured data are averaged over the four measured data, as shown in Fig. 10. The ground test result shows the

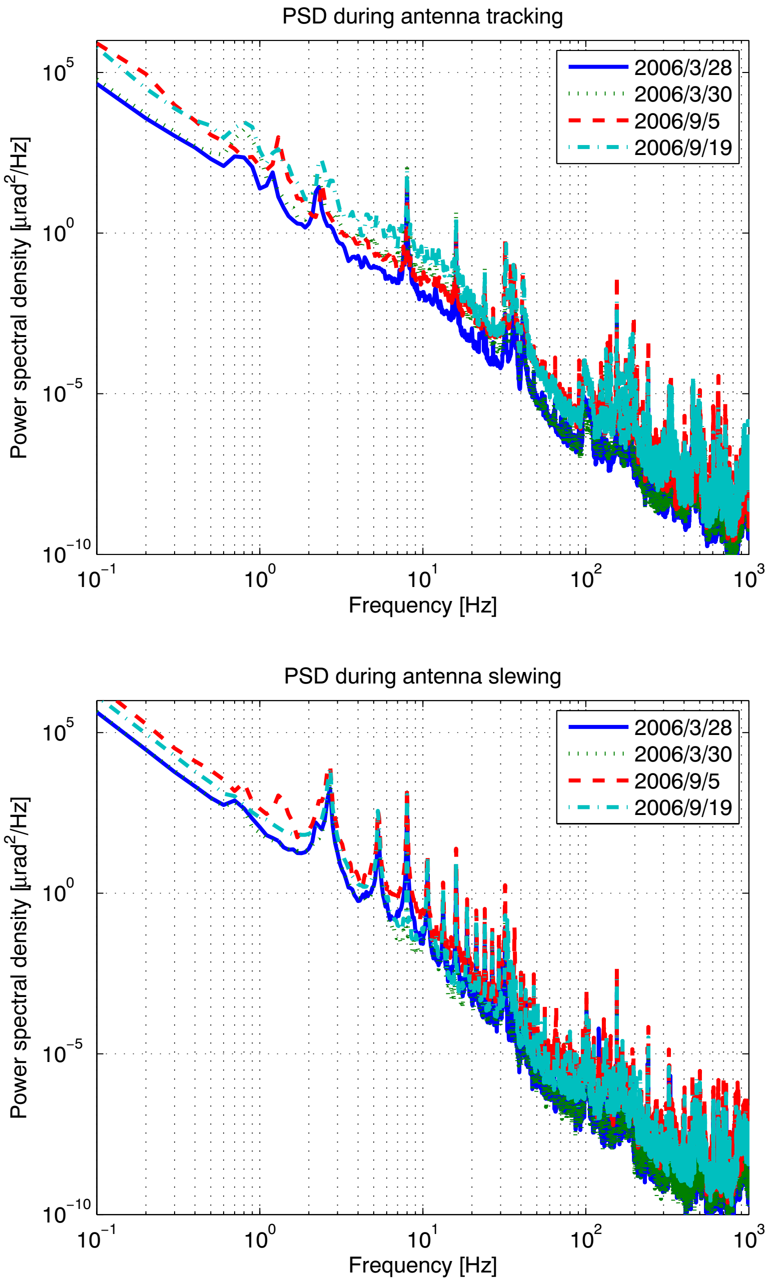

Fig. 9 PSDs of the angular microvibrations while the optical antenna is performing (a) the tracking operation and (b) the slewing operation.

worst-case PSD, which was experimentally measured in the laboratory. The in-orbit measured data are generally lower than the ground test results. As shown in Fig. 11, the microvibration test on the ground is an effective method for the verification of the microvibrational noise in the satellite systems in orbit.

Figure 12 shows the comparison among the PSDs measured at several space demonstrations in orbit. The PSDs for antenna tracking and slewing are derived as the worst-case power spectra during the experiments. The measured PSDs for the OICETS satellite show almost the same spectral levels as compared to those obtained from other data. The PSD from LANDSAT had frequency components up to 128 Hz. The in-orbit measurement onboard OLYMPUS was done by using the same kind of accelerometers, but the frequency components were [5,6] up to about 500 Hz. The other in-orbit measurement onboard ETS-VI was done by using the angular displacement of fine steering mirrors for the laser communications, which was[13,18] up to 250 Hz. In this paper, the in-orbit measurements of the angular microvibration from 1 to 1024 Hz are additionally provided by using accelerometers as a database of the real measurement results. The differences in spectral peaks among each measured data depend on the material and structure of the satellite systems. The measured spectra are considered to have reasonable values as compared with other published data.

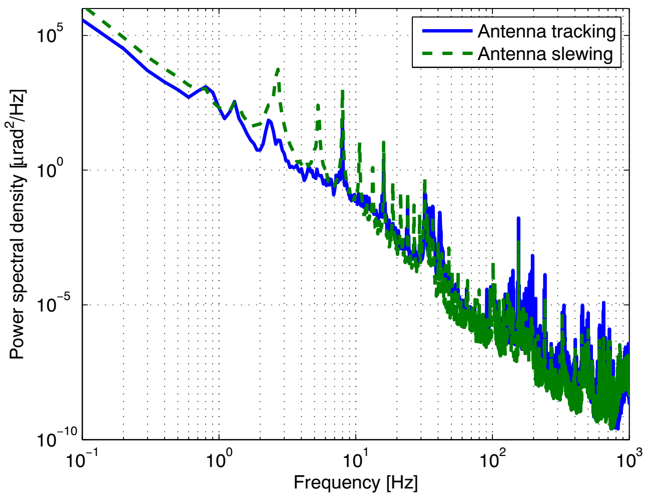

Fig. 10 PSDs of the angular microvibrations while the optical antenna is in the tracking and slewing modes, respectively.

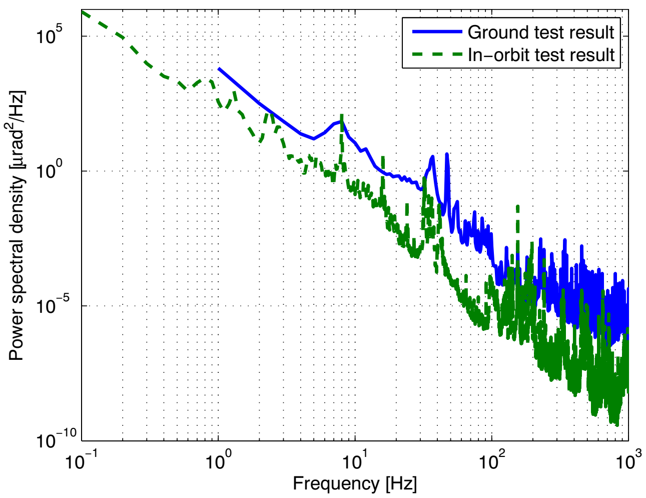

Fig. 11 Comparison between the PSDs during tracking measured in the ground test and in space.

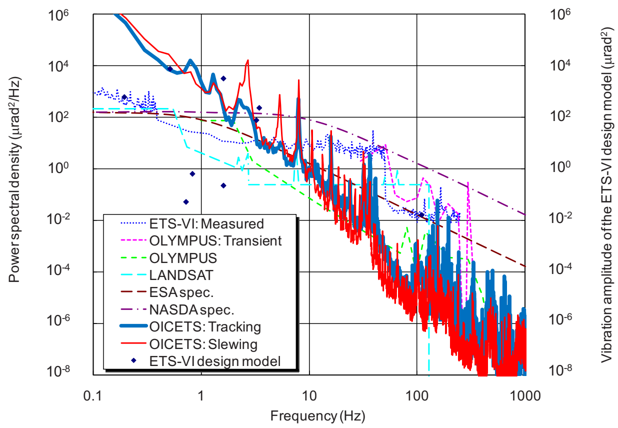

Fig. 12 Comparison between the PSDs measured in space for several space demonstrations.

The rms value of the spacecraft base motion at the interface point of the payload is expressed by

$$\Phi_{\mathrm{rms}} = \left[\int \theta^2_{\mathrm{rms}}(f)\mathrm{d}f \right]^{1/2}, \tag{17}$$

which is estimated from the PSD. The rms angular disturbance from 1 to 1024 Hz is estimated to be 23.8 μrad rms during tracking and 43.8 μrad rms during slewing, for the worst cases, respectively. In the ground test,[18] the rms value of the spacecraft base motion at the interface point of the payload was 85.3 μrad rms, which was estimated from the PSD for 1 to 2048 Hz, as the worst-case condition that includes microvibrations from the laser terminal itself. Table 2 shows the frequency components of the PSDs from 1 to 1024 Hz for antenna tracking and slewing, respectively. The frequency components almost reside in frequencies lower than 10 Hz.

It can be predicted that the in-orbit microvibrations will be greater, because during the ground tests a part of the microvibrational energy is dissipated into the air as heat or sound. Such energy dissipation does not occur in the space environment; however, it is also considered that the contribution of acoustical noise in the ground test might not be negligible because acoustical noise does not exist in orbit. These effects might be included in the in-orbit measured results.

Table 2 Frequency components of the worst-case rms angular disturbance during antenna tracking and slewing.

Frequency Band	1–10 Hz	10–100 Hz	100–1024 Hz	Total
rms disturbance during	22.0	1.6	0.1	23.8
tracking in μrad rms (%)	(92.8%)	(6.9%)	(0.3%)	(100%)
rms disturbance during	41.5	2.2	0.0	43.8
slewing in μrad rms (%)	(94.9%)	(5.1%)	(0%)	(100%)

7 Conclusion

Microvibrations on the OICETS satellite were measured in orbit and analyzed using data acquired during the ground-to-satellite laser communications experiments. The acceleration level measured by the onboard accelerometers reached approximately 0.5 m/s^2 ($=50$ mG). At the interface point of the laser communications payload, the acceleration level was estimated to be approximately 0.1 m/s^2 ($=10$ mG) during the experiments. The worst-case rms values of the angular microvibrational disturbance on the spacecraft was estimated to be 23.8 and 43.8 μrad rms from 1 to 1024 Hz, while the optical antenna was in the tracking and slewing modes, respectively. Optical tracking by the laser communication terminal was maintained under these microvibrational conditions, which adequately satisfied the requirements for the optical tracking performance in orbit.

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